JORDAN
INVESTMENT COMMISSION
الأردن
هيئة الاستثمار

A pre-feasibility study for the establishment of medicinal herbs plant in Ma'an


2017

Table of contents

List of Tables

Table of Figures

1. Executive summary

The project aims to establish a herbs and medicinal plants factory in Ma'an Development Area, while encouraging farmers in Ma'an on the cultivation of herbs to purchase crops from them and from the rest of the governorates, wherebyThe shortage of herbs and medicinal plants will be imported from neighboring countries.

In addition to determining the types of rare plants that will be used in the area, where Ma'an has the advantage of having large quantities of the desired plants for aromatic and medicinal uses. The plant will extract the aromatic and medicinal oils from these plants then market them, after filling themin custom packges to fulfill Jordan's need of herbal extracts and the opportunity to export the products abroad due to the growing demand.

Table 1 economical indicators

Indicator	
Total Investment Cost (Jordan Dinars)	1,598,203
Internal Rate of Return (IRR)	27.85%
Pay-Back Period (Years)	4
Net Present Value (Jordan Dinars)	1,743,799
Job Opportunities provided by the Project	25
Total Revenues (For 10 Years)	29,091,150
Total Operating Expenses (For 10 years)	21,658,379
Total Net Profit (For 10 years)	3,574,103

Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

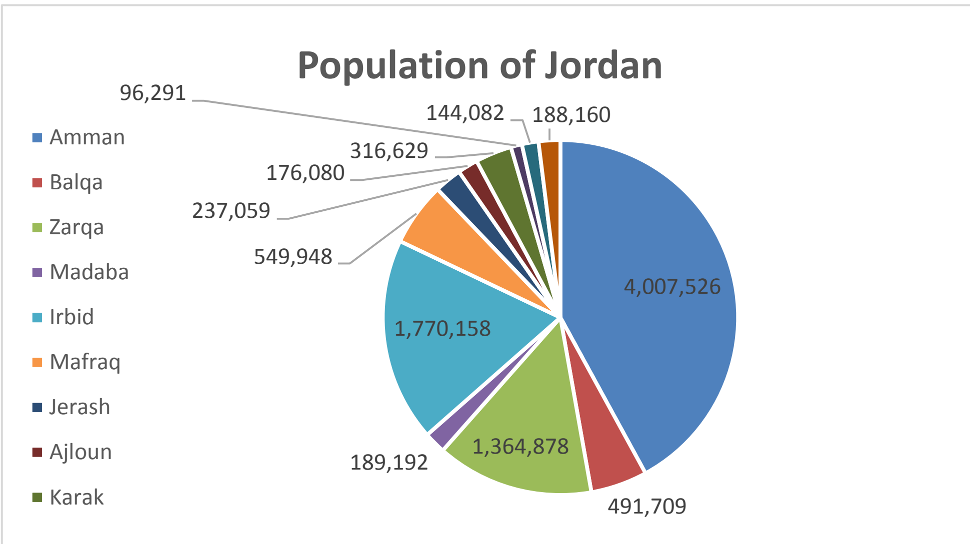
Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the

Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 2 Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	**%18.2**

15-64 Years	%61.02	%30.7	**%28.6**
More than 65 Years	%3.94	%1.6	**%1.6**

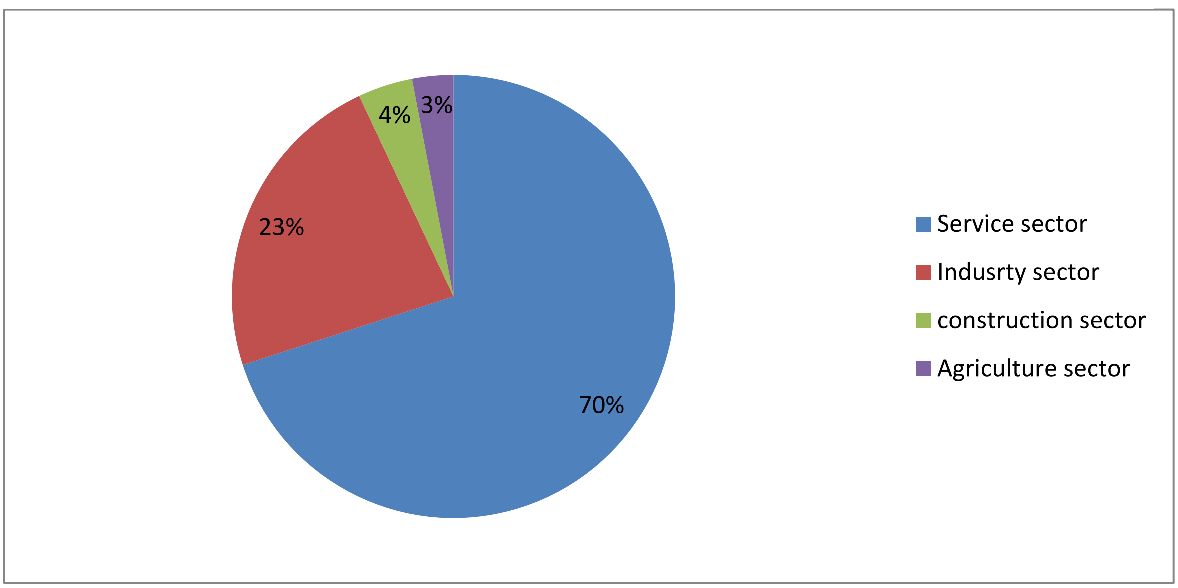
Department of Statistics (DOS), 2015.

Overiew of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2 Sectors Contibution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US$3.8 trillion market and compromising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.
Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.
Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.
Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.
The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in it endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Invesment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.

- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

The investment environment in Ma'an:

As per in the Investment Law No. 30 of 2014 and the Income Tax Reduction Law in the Less Developed Areas (issued under Article (5) of the Investment Law No. (30) Of 2014), the following taxes and exemptions are imposed on investment projects:

Table 3 payable income tax and tax exemptions

Development areas	Less developed areas
Income tax is 5%	Value of income tax 20% According to the divisions in the law, Ma'an governorate is classified as category B where investors get 80% tax exemption on income tax for 20 years from the date of investment.

2. Market Analysis

Project Description

The project aims to establish a herbs and medicinal plants factory in Ma'an Development Area, while encouraging farmers in Ma'an on the cultivation of herbs to purchase crops from them and from the rest of the governorates, wherebyThe shortage of herbs and medicinal plants will be imported from neighboring countries.

In addition to determining the types of rare plants that will be used in the area, where Ma'an has the advantage of having large quantities of the desired plants for aromatic and medicinal uses. The plant will extract the aromatic and medicinal oils from these plants then market them, after filling themin custom packges to fulfill Jordan's need of herbal extracts and the opportunity to export the products abroad due to the growing demand.

Description of the Production Process

The idea of the project is to buy medicinal herbs from Ma'an and other provinces and importing others from neighboring countries.

The cost of raw materials (herbs and medicinal plants) that are purchased and imported needed for the production process equals around 60% of the value of the total revenue on the basis of cost of raw materials and the cost of imprting them.

Production Stages

• Gathering herbs and medicinal plants in bags and moving them to the warehouse

• Washing herbs using machines specific for washing herbs and plants

• Drying the herbs and adjusting the temperature to maintain volatile oils and biomaterials

• Grinding the herbs by using milling machines

• Extracting useful raw materials with therapeutic benefits from the herbs

•Packaging

Project Objectives

The objectives of the project are as follows:

- Achieve profitable returns for those in charge of the project
- Hire people and reduce unemployment rates
- Provide targeted markets with their need of medicinal herbs
- Exploit the vast agricultural lands in the Governorate of Ma'an
- Contribute in the development of the agricultural and industrial sector and provide jobs

Proposed Product

The project will fill medicinal herbs (chamomile, fennel, thyme, sage, anise, ginger) in cloth bags (tea bags), then package them in carton packaging. Based on information available from the manufacturers of tea bags, it was assumed that the weight per bag is (8.1 to 5.2) grams and each package consists of (100-110) bags.

Project Justifications

Global markets experiencing significant growth in demand for medicinal herbs consumption, noting that medicinal herbs of various kinds are available in Jordan and therefore, most of the proposed project revenues will be local, in addition to the opportunity for export.

Target segmants

- Local market
- International market

Market Size Analysis

Market size analysiswas conducted according to exports and imports in Jordan market as shown in Table (4)

Table 3 Summary of Annual Needs of Required Herbs

Category Description	Year	Imports: Jordanian Dinars	Imports: (KG)	National exports: Jordanian Dinars	National exports: (KG)	Re-exported: JD	Re-exported: (KG)	Total exports: Jordanian Dinars
Green tea (not fermented) in containers are ready-to-trade and weighing less than or equal to (3) kg	2015	639,116	64,960	2,292,159	444,046	10,635	4,000	2,302,794
Green parsley, fresh or chilled	2015	0	0	6,153	38,900	0	0	6,153
Coriander seeds, crushed or not	2015	943,715	1,193,809	0	0	133,830	168,100	133,830
Seeds of anise, crushed	2015	766,966	422,163	0	0	44,738	27,000	44,738
Thyme, bay leaves (Rand), Curry	2015	4,229,651	1,272,948	4,481,916	3,241,281	171,264	139,016	4,653,180
Crushed Ginger	2015	1,127,707	934,972	0	0	145,507	98,697	145,507

Source: Department of Statistics - economic statistics (2015)

- **1.1 Competitors Analysis**

The number of competitors in the domestic market of herbal tea factories are 10 factories.

Examples of direct competitors:

• Nutri Dar PLC.

• Narain Food Industries / Mega Food Industries.

• Salsabil specialized industries.

• Jordan's lush natural cosmetics.

• Shams Al Marefa of the trade and industry of dietary supplements.

• Al Arrab of natural cosmetics.

Production Capacity

Production quantity of natural herbal plants has been calculated by assuming working 330 work days per year and 9 hours of work per day. The production capacity estimated by 800 packets\ hour as follows:

Production capacity = 800x 9 x 330 = 2,376,000 packets \ years.

Table 4 Production capacity

Analysis of Production Capacity	
Production capacity of the project (packets)	2376000
The proportion of Chamomile production from the overall production	15.000%
The proportion of fennel production from the overall production	15.000%
The proportion of thyme production from the overall production	25.000%
The proportion of Sagebrush production from the overall production	15.000%
The proportion of of anise production from the overall production	15.000%
The proportion of ginger production from the overall production	15.000%

Price Analysis and Pricing Policy

Table 5 Prices of the Required Raw Materials for Manufacturing

Required raw materials	Quantity	Cost (JD)
chamomile	1 ton	5000
Fennel	1 ton	2000

Required raw materials	Quantity	Cost (JD)
Thyme	1 ton	1500
Sage	1 ton	4000
Anise	1 ton	3000
Ginger	1 ton	6000
Preservatives	1 ton	1773

Table 6 the proposed selling prices per packets (200 gm)

Prices	JD
Chamomile	2
Fennel	0.85
Breckland Thyme	1.1
Sage	2.5
Anise	1.2
Ginger	2.5

Projected Revenues

Table 7 project revenues during the years 2018 to 2027

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Production rate	60%	60%	75%	75%	75%	80%	80%	80%	80%	85%
Chamomile revenues	427680	427680	534600	534600	534600	570240	570240	570240	570240	605880
Fennel revenues	181764.0	181764.0	227205.0	227205.0	227205.0	242352.0	242352.0	242352.0	242352.0	257499.0
Wild thyme revenues	392040	392040	490050	490050	490050	522720	522720	522720	522720	555390
Sagebrush revenues	534600	534600	668250	668250	668250	712800	712800	712800	712800	757350
Anise revenues	256608	256608	320760	320760	320760	342144	342144	342144	342144	363528
Ginger revenues	534600.0	534600.0	668250.0	668250.0	668250.0	712800.0	712800.0	712800.0	712800.0	757350.0
Tota revenues	**2327292**	**2327292**	**2909115**	**2909115**	**2909115**	**3103056**	**3103056**	**3103056**	**3103056**	**3296997**

3. Technical Analysis

Required Human Resources

Table 8 the required labor for the project

Job title	No.
Indirect staff	
General Manager	1
Financial Manager	1
Production Manager	1
Accountants	2
Director of Human Resources	1
Marketing Representative	3
Warehouse security guard	1
Secretary	1
Driver	2
Guard	1
Total Non-Direct staff	14
Chemical engineer	2
Pharmacist assistant and lab technician	1
Technicians Workers	8
Total Direct staff	11
total summation	25

• **General Manager:** will perform all the tasks and responsibilities of the general manager, and have active participation in the identification and formulation of objectives, as well as planning and organizing workflow of the facility to ensure the achievement of objectives, and alignment with professional safety and security regulations.

• **Financial Manager**: will be responsible for the implementation of the plans, regulations, administrative, accounting and financial procedures, as well as monitoring and controlling the resources and expenditures of the department, and the organization of the budget and activities in the department. Also, prepare financial reports.

• **Production Manager**: will be responsible for the technicaland production matters and oversee the production process.

• **Human Resources Director**: will be responsible for planning, recruitment of human resources for the company and work to measure their performance and develop their abilities through effective training programs and work to stimulate employees to be creativity and innovative in the areas of their business and work to regulate the accumulation of experience and knowledge in the areas of work.

• **Laboratory technician:** will be responsible for the organization of work within the laboratories and ensure the required quality.

• **Marketing officer**: will study the market needs, and follow-up with advertising and media outlets and assess the effectiveness of activities, along with monitoring the level of sales.

• **Accountant**: will be responsible for the implementation of financial and accounting procedures in the plant.

• **Packaging technician:** will be responsible for packaging and arranging products in sterilized packages and prepare them for sale.

• **Guard:** will maintain the security of the plant.

• **Maintenance Technician**: will be responsible all the electrical and mechanical maintenance, and follow up on the implementation of plans and programs in order to ensure the continuity of the production process and quality at full capacity with the lowest possible cost, and make sure of the availability of spare parts, and the preparation of technical reports.

• **Secretary**: will respond to calls and arrange meetings for managers.

• **Public Safety staff**: will study potential risks in the factory and work to provide a safe working environment.

• **Development Department staff**: will develop the efficiency and product quality and reduce costs of production.

• **Quality department staff**: will perform quality control over the product by reducing defects, and improve production.

• **Hygiene staff**: will maintain the cleanliness of the plant and offices.

• **Driver**: will deliver products to customers.

Site Analysis

The idea of the project aims to exploit the land in Ma'an and encourage farmers to grow herbs and medicinal plants; since Maan area is suitable for agriculture by using modern agricultural methods, as well as creating jobs and improving the economic situation. Add to that the price of land in Ma'an Development Area is low compared to the price of industrial land prices in other regions of the Kingdom,In addition to the features and exemptions obtained by the region. The MDA is distinguished by the following:

• Located near Saudi Arabia and on the main road that connects the south of Jordan and Iraq, allowing investors to benefit from a bigger market

• Constitutes a major component of a transport and storage system for exports and imports of the Hashemite Kingdom of Jordan through its proximity to the port of Aqaba at a distance of 110 km and its proximity to the land allocated for the land port and the existing railway

Technical Requirements Analysis

1.1.1 Land and Construction

The laws and regulations of Ma'an Development Area states that the lowest space for the plant must be 4 Donms, including 2 Donms for construction work. The project will be constructed on an area of 4000 sq.m.

The following table shows the needed land:

Table9 Required Land for the Project

Land	Cost (JD/sq.m)	Area sq.m
The required land	13	**4000**

Source: Land registration – Ma'an

The following table shows the required construction work for the project:

Table 10 Required Construction Work

Construction work	Area (m²)
Buildings	2,000
Asphalt works	500

Machinery and Equipment

The following table shows the required machinery and equipment needed for the project:

Table 11 Machinery and equipments

Machinery and equipment	Quantity	Cost (JD per unit)
Washing quipment	2	8,000
Driers	2	6,000
Milling equipment	2	16,000
Juices extracting equipment	2	50,000
Packaging line	1	60,000
Mixing and blending machine	1	40,000
Laboratory Equipments	1	25,000
Forklift	2	12,000

Vehicles and Transportation

The following table shows a list of vehicles and meanss of transportion required for the project

Table 12 Vehicles and modes of transportation

Vehicles and modes of transportation	Quantity	Unit cost (JD)
Pickup truck	2	23,400

4. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 12.05% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Icome tax on the net project income is 5 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 13 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 14 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 15 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	1.00%
Annual Expenses Increase Rate	2.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 16 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	60.00%
Utilities Expenses from Total Revenues	2.5%
Maintenance Expenses from Total Revenues	1.00%
Depreciation Expenses from Total Revenues	0.50%
Insurance Expenses from Total Assets Value	1.00%
Marketing Expenses from Total Revenues	1.00%

Table 17 Income Tax Assumptions

Income Tax Assumptions	Value
Average Income Tax in Jordan(1)	5%
Income Tax Deduction(2)	0%
Income Tax after Deduction	5%
Compulsary Reserve Percentage(3)	10%
Other Assumptions	**Value**
Annual Monthly Salaries have been calculated after multiplying monthly salaries	16

Risk Premuim	
Risk Free Rate of Return (1)	6.50%
Return to Debt Maturity (2)	9.48%
Market Risk Premuim (3)	10.12%
Income Tax Rate	5.00%
Beta (4)	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 18 : Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premuim	%10.12
Income Tax Rate	%5
Bets	1.00
Equity	639,281
Loans Value	958,922
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%9.01
Loans Value	958,922
Loans Percentage	%60
Equity	
Cost of Equity	%16.6
Equity Value	639,281
Equety Percentage	%40
Gross	
Project Value	1,598,203
Proportion of the project	%100
Weighted Average Cost of Capital	%12.05

Capital Expenditures

The estimated cost of the project is 1,598,302 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 19 Land Capital Expenditure

Land	(Jordan Dinar/Meters Square) Cost	Area (M2)	Cost
Required Land	13	4,000	52,000
Cost			**52,000**

Table 20 Construction Work Capital Expenditure

Description	Cost (JOD per Squared Meters)	Area (Squared Meters)	Gross Cost Jordan /
Buildings	250	2,000	500,000
Paving	12	500	6,000
Other	0	-	25,000
Total			**531,000**

Table 21 Machinery and Equipment Capital Expenditure

Machinery and equipment	Quantity	Cost / JD	Total Cost
Washing quipment	2	8,000	16,000
Driers	2	6,000	12,000
Milling equipment	2	16,000	32,000
Juices extracting equipment	2	50,000	100,000
Packaging line	1	60,000	60,000
Mixing and blending machine	1	40,000	40,000
Laboratory Equipments	1	25,000	25,000
Forklift	2	12,000	24,000
Total (JD)			**309,000**

Table 22 Furniture Capital Expenditure

Furniture and furnishings	Quantity	Cost / JD	Total Cost
Furniture	-	-	20,000
Total			20,000

Table 23 Transporation and Vehicles Capex

Transporation and Vehicles	Quantity	Unit Cost/JOD	Total Cost/JOD
Pick up	2	23,400	46,800
Total			**46,800**

Table 24 Pre Operating Expenses

Pre-operating expenses	Estimaited cost (JD)
Governmental fees	1,000
Transportation	1,200
Marketing	2,400
Training	600
Total	**5,200**

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 25 Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	3	%25	434,023
General and administrative expenses	3	%25	5,870
Marketing expenses	3	%25	5,818
Indirect Salaries	3	%25	43,200
Total			**488,912**

Table 26 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Land	52,000	10%	57,200	%3.6
Construction works	531,000	10%	584,100	%36.5
Machinery &Equipment	309,000	10%	339,900	%21.3
Furniture	20,000	10%	22,000	%1.4
Vehicles	46,800	10%	51,480	%3.2
Pre-Operating Expenses	5,200	10%	5,720	%0.4

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Working capital	488,912	10%	537,803	%33.7
Total (JD)	**1,452,912**		**1,598,203**	**100%**

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (1,598,203) JD, where 60% of the project will be financed through loans (958,922) JD, and the remaining (40%) through (Equity) with (639,281) JD. The following table summarizes the general structure of the required financing:

Table 27 Sources of Funding

Funding Sources	Amount	Percentage
Equity (Self Financing)	639,281	%40
Loans	958,922	%60
Total	**1,598,203**	**%100**
Use of Fund	Value	Ratio
Capital Expenditures	1,054,680	%65.99
Pre-Operating Expenses	5,720	%0.36
Working Capital	537,803	%33.65
Total	**1,598,203**	**%100.00**

Operating Expenses

The following table sumarizes the expected operating expenses for the proposed project (2018-2027):

Table 28 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials	1,396,375	1,396,375	1,745,469	1,745,469	1,745,469	1,861,834	1,861,834	1,861,834	1,861,834	1,978,198
Utilities from	58,182	58,182	72,728	72,728	72,728	77,576	77,576	77,576	77,576	82,425
Maintenance from	23,273	23,273	29,091	29,091	29,091	31,031	31,031	31,031	31,031	32,970
Disposables from	11,636	11,636	14,546	14,546	14,546	15,515	15,515	15,515	15,515	16,485
Direct Salaries	88,800	92,441	96,231	100,176	104,284	108,559	113,010	117,644	122,467	127,488
Others	157,827	158,191	195,806	196,201	196,612	209,452	209,897	210,360	210,842	223,757
Total	**1,736,094**	**1,740,098**	**2,153,871**	**2,158,211**	**2,162,729**	**2,303,967**	**2,308,863**	**2,313,959**	**2,319,265**	**2,461,323**

General and Administrative Expenses

The following table sumarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 29 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	2,400	2,448	2,497	2,547	2,598	2,650	2,703	2,757	2,812	2,868
Hospitality	1,200	1,224	1,248	1,273	1,299	1,325	1,351	1,378	1,406	1,434

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Staionary and Printing	1,200	1,224	1,248	1,273	1,299	1,325	1,351	1,378	1,406	1,434
Transportaton	4,800	4,896	4,994	5,094	5,196	5,300	5,406	5,514	5,624	5,736
Training	1,200	1,224	1,248	1,273	1,299	1,325	1,351	1,378	1,406	1,434
Insurance	10,547	10,758	10,973	11,192	11,416	11,645	11,877	12,115	12,357	12,604
Miscellaneous	2,135	2,177	2,221	2,265	2,311	2,357	2,404	2,452	2,501	2,551
Total	**23,481**	**23,951**	**24,430**	**24,919**	**25,417**	**25,925**	**26,444**	**26,973**	**27,512**	**28,063**

Marketing Expenses

Table 30 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	23,273	23,273	29,091	29,091	29,091	31,031	31,031	31,031	31,031	32,970

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 31 Expecte

d staff preparation

Job Title	No.	Monthly salaries	Total monthly salaries	Total yearly salaries
General Manager	1	2,500	2,500	40,000
Financial Manager	1	1,500	1,500	24,000
Production Manager	1	1,200	1,200	19,200
Accountants	2	450	900	14,400
Human Resources Manager	1	1,200	1,200	19,200
Marketing Representative	3	450	1,350	21,600
Warehouse security guard	1	650	650	10,400
Secretary	1	450	450	7,200
Driver	2	350	700	11,200
guard	1	350	350	5,600
Total Indirect staff	**14**		**10,800**	**172,800**
Chemical Engineer	2	850	1,700	27,200
Pharmacist assistant and lab technician	1	650	650	10,400
Technicians Workers	8	400	3,200	51,200
Total direct staff	**11**		**5,550**	**88,800**
Grand Total	**25**		**16,350**	**261,600**

Table 32 Expected monthly salaries (2018-2027)

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect staff										
General Manager	2,500	2,603	2,709	2,820	2,936	3,056	3,182	3,312	3,448	3,589
Financial Manager	1,500	1,562	1,626	1,692	1,762	1,834	1,909	1,987	2,069	2,154
Production Manager	1,200	1,249	1,300	1,354	1,409	1,467	1,527	1,590	1,655	1,723
Accountants	450	468	488	508	528	550	573	596	621	646
Human Resources Manager	1,200	1,249	1,300	1,354	1,409	1,467	1,527	1,590	1,655	1,723
Marketing Representative	450	468	488	508	528	550	573	596	621	646
Warehouse security guard	650	677	704	733	763	795	827	861	896	933
Secretary	450	468	488	508	528	550	573	596	621	646
Driver	350	364	379	395	411	428	445	464	483	502
Guard	350	364	379	395	411	428	445	464	483	502
Direct staff										
Chemical Engineer	850	885	921	959	998	1,039	1,082	1,126	1,172	1,220
Pharmacist assistant and lab technician	650	677	704	733	763	795	827	861	896	933
Technicians Workers	400	416	433	451	470	489	509	530	552	574

Table 31 Annual Expected Salaries

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect staff										

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
General Manager	40,000	41,640	43,347	45,124	46,975	48,901	50,905	52,993	55,165	57,427
Financial Manager	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Production Manager	19,200	19,987	20,807	21,660	22,548	23,472	24,435	25,436	26,479	27,565
Accountants	14,400	14,990	15,605	16,245	16,911	17,604	18,326	19,077	19,860	20,674
Human Resources Manager	19,200	19,987	20,807	21,660	22,548	23,472	24,435	25,436	26,479	27,565
Marketing Representative	21,600	22,486	23,408	24,367	25,366	26,406	27,489	28,616	29,789	31,011
Warehouse security guard	10,400	10,826	11,270	11,732	12,213	12,714	13,235	13,778	14,343	14,931
Secretary	7,200	7,495	7,803	8,122	8,455	8,802	9,163	9,539	9,930	10,337
Driver	11,200	11,659	12,137	12,635	13,153	13,692	14,254	14,838	15,446	16,080
Guard	5,600	5,830	6,069	6,317	6,576	6,846	7,127	7,419	7,723	8,040
Total Indirect	**172,800**	**179,885**	**187,260**	**194,938**	**202,930**	**211,250**	**219,912**	**228,928**	**238,314**	**248,085**
Direct staff										
Chemical engineer	27,200	28,315	29,476	30,685	31,943	33,252	34,616	36,035	37,512	39,050
Pharmacist assistant and lab Technician Workers	10,400	10,826	11,270	11,732	12,213	12,714	13,235	13,778	14,343	14,931
Technicians	51,200	53,299	55,484	57,759	60,127	62,593	65,159	67,831	70,612	73,507
Total Direct	**88,800**	**92,441**	**96,231**	**100,176**	**104,284**	**108,559**	**113,010**	**117,644**	**122,467**	**127,488**
Grand Total	**261,600**	**272,326**	**283,491**	**295,114**	**307,214**	**319,810**	**332,922**	**346,572**	**360,781**	**375,573**

Depreciations

The following tables illustrate cost of Capex and Depreciaition rates:

Table 33 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	57,200	%0.0	%0.0
Construction Works	584,100	%5.0	%0.0
Machinaries	339,900	%10.0	%2.0
Furniture	22,000	%10.0	%5.0
Vehicles	51,480	%10.0	%0.0
	1,054,680		

Table 34 Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciaitions	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	1,054,680	1,062,578	1,070,667	1,078,952	1,087,440	1,096,135	1,105,045	1,114,174	1,123,531	1,133,121
Total Depreciation Expenses	70,543	71,333	72,142	72,970	73,819	74,689	75,579	76,492	77,428	78,387
Total Accumulated Depreciation	70,543	141,876	214,017	286,988	360,807	435,495	511,075	587,567	664,995	743,382
Total Additions	0	7,898	8,089	8,285	8,487	8,695	8,909	9,130	9,357	9,590
Total Net Book Values	984,137	920,702	856,649	791,965	726,633	660,640	593,970	526,607	458,536	389,739

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 35 Loan Details

Loan Value	958,922
Annual Interst Rate	9.48%
Loans Period	5
Loan Starts at year:	2018
Annual Payment	249,610
Number of Payments	5

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				958,922
2019	249,610	90,906	158,704	800,218
2020	249,610	75,861	173,749	626,469
2021	249,610	59,389	190,220	436,249
2022	249,610	41,356	208,253	227,996
2023	249,610	21,614	227,996	0

Income Statement

Table 36 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	2,327,292	2,327,292	2,909,115	2,909,115	2,909,115	3,103,056	3,103,056	3,103,056	3,103,056	3,296,997
Gross Operating Revenues	2,327,292	2,327,292	2,909,115	2,909,115	2,909,115	3,103,056	3,103,056	3,103,056	3,103,056	3,296,997
Operating Expenses	-1,736,094	-1,740,098	-2,153,871	-2,158,211	-2,162,729	-2,303,967	-2,308,863	-2,313,959	-2,319,265	-2,461,323
Gross Operating Profit	591,198	587,194	755,244	750,904	746,386	799,089	794,193	789,097	783,791	835,674
Gross Profit Percentage	*%25*	*%25*	*%26*	*%26*	*%26*	*%26*	*%26*	*%25*	*%25*	*%25*
Salaries and Benefits (Indirect Staff)	-172,800	-179,885	-187,260	-194,938	-202,930	-211,250	-219,912	-228,928	-238,314	-248,085
General and Administraive Expenses	-23,481	-23,951	-24,430	-24,919	-25,417	-25,925	-26,444	-26,973	-27,512	-28,063
Markeing Expenses	-23,273	-23,273	-29,091	-29,091	-29,091	-31,031	-31,031	-31,031	-31,031	-32,970
Pre Operating Expenses	-5,720									
Gross Indirect Expenses	-225,274	-227,109	-240,781	-248,948	-257,438	-268,206	-277,386	-286,931	-296,857	-309,117
Income before Interest, Depreciation and Tax	365,924	360,085	514,463	501,956	488,948	530,883	516,807	502,165	486,934	526,557
Fixed Assets Depreciations	-70,543	-71,333	-72,142	-72,970	-73,819	-74,689	-75,579	-76,492	-77,428	-78,387
Income before Tax and Interests	295,381	288,752	442,321	428,986	415,129	456,195	441,228	425,673	409,506	448,170
Bank Interests	-90,906	-75,861	-59,389	-41,356	-21,614	0	0	0	0	0

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Income Before Tax	204,475	212,891	382,932	387,630	393,515	456,195	441,228	425,673	409,506	448,170
Income Tax	-10,224	-10,645	-19,147	-19,381	-19,676	-22,810	-22,061	-21,284	-20,475	-22,408
Net Profit	194,251	202,247	363,785	368,248	373,839	433,385	419,166	404,389	389,031	425,761
Net Profit Percentage	*%8*	*%9*	*%13*	*%13*	*%13*	*%14*	*%14*	*%13*	*%13*	*%13*
Compulsory Reserves	-19,425	-20,225	-36,379	-36,825	-37,384	-43,338	-41,917	-40,439	-38,903	-42,576
Retained Earnings	174,826	356,848	684,255	1,015,678	1,352,134	1,742,180	2,119,430	2,483,380	2,833,508	3,216,693

Expected Cashflows Statement

The table belwo summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 37 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash flows from operating activities										
Net profit (loss)	194,251	202,247	363,785	368,248	373,839	433,385	419,166	404,389	389,031	425,761
Non-monetary adjustments										
Bank benefits	90,906	75,861	59,389	41,356	21,614	0	0	0	0	0
Consumption	70,543	71,333	72,142	72,970	73,819	74,689	75,579	76,492	77,428	78,387
Operating cash flows before changes in working capital	355,700	349,440	495,316	482,575	469,272	508,073	494,746	480,882	466,459	504,148

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Shortage (increase) in inventory	-116,365	0	-29,091	0	0	-9,697	0	0	0	-9,697
Shortage (increase) in receivables	-193,941	0	-48,485	0	0	-16,162	0	0	0	-16,162
(Decrease) increase in accounts payable	84,930	1,001	16,170	1,085	1,129	6,218	1,224	1,274	1,326	6,423
Change in working capital	-225,376	1,001	-61,407	1,085	1,129	-19,641	1,224	1,274	1,326	-19,436
Net cash flows (used) from operating activities	130,324	350,441	433,909	483,660	470,401	488,433	495,970	482,156	467,785	484,713
Cash flows from investing activities										
(Purchase) of fixed assets	-1,054,680	-7,898	-8,089	-8,285	-8,487	-8,695	-8,909	-9,130	-9,357	-9,590
Net flows (used) from investment activities	-1,054,680	-7,898	-8,089	-8,285	-8,487	-8,695	-8,909	-9,130	-9,357	-9,590
Cash flows from financing activities										
Equity capital	639,281									
Loan Amortization	-158,704	-173,749	-190,220	-208,253	-227,996	0	0	0	0	0
Bank Interest Rate	-90,906	-75,861	-59,389	-41,356	-21,614	0	0	0	0	0
Loans	958,922	0	0	0	0	0	0	0	0	0
Net cash flows (used) from financing activities	1,348,594	-249,610	-249,610	-249,610	-249,610	0	0	0	0	0
Net increase (decrease) in cash	424,238	92,934	176,211	225,765	212,304	479,737	487,060	473,026	458,429	475,123
Cashflows at the Beginning of Period	0	424,238	517,172	693,382	919,147	1,131,452	1,611,189	2,098,250	2,571,276	3,029,704

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cashflows at the End of Period	424,238	517,172	693,382	919,147	1,131,452	1,611,189	2,098,250	2,571,276	3,029,704	3,504,827

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 38 Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current assets										
Fund	424,238	517,172	693,382	919,147	1,131,452	1,611,189	2,098,250	2,571,276	3,029,704	3,504,827
Inventory	116,365	116,365	145,456	145,456	145,456	155,153	155,153	155,153	155,153	164,850
Accounts receivable	193,941	193,941	242,426	242,426	242,426	258,588	258,588	258,588	258,588	274,750
Total assets traded	**734,543**	**827,477**	**1,081,264**	**1,307,029**	**1,519,334**	**2,024,930**	**2,511,990**	**2,985,017**	**3,443,445**	**3,944,427**
Non-current assets										
Fixed assets (net)	984,137	920,702	856,649	791,965	726,633	660,640	593,970	526,607	458,536	389,739
Total non-current assets	**984,137**	**920,702**	**856,649**	**791,965**	**726,633**	**660,640**	**593,970**	**526,607**	**458,536**	**389,739**
Total assets	**1,718,680**	**1,748,179**	**1,937,914**	**2,098,994**	**2,245,967**	**2,685,570**	**3,105,960**	**3,511,624**	**3,901,981**	**4,334,166**
Liabilities	-	-	-	-	-					
Current liabilities										

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Payables	84,930	85,931	102,100	103,185	104,315	110,533	111,757	113,031	114,358	120,781
Remaining amount of Loan	173,749	190,220	208,253	227,996	0	0	0	0	0	0
Total current liabilities	258,679	276,151	310,354	331,181	104,315	110,533	111,757	113,031	114,358	120,781
Non-current liabilities										
long term loans	626,469	436,249	227,996	0	0	0	0	0	0	0
Total long - term liabilities	626,469	436,249	227,996	0	0	0	0	0	0	0
Total liabilities	885,148	712,400	538,349	331,181	104,315	110,533	111,757	113,031	114,358	120,781
Property rights										
Member Contributions	639,281	639,281	639,281	639,281	639,281	639,281	639,281	639,281	639,281	639,281
Compulsory reserve	19,425	39,650	76,028	112,853	150,237	193,576	235,492	275,931	314,834	357,410
Profit (loss) round	174,826	356,848	684,255	1,015,678	1,352,134	1,742,180	2,119,430	2,483,380	2,833,508	3,216,693
Total equity	833,533	1,035,779	1,399,565	1,767,813	2,141,652	2,575,037	2,994,203	3,398,592	3,787,623	4,213,384
Total liabilities and equity	1,718,680	1,748,179	1,937,914	2,098,994	2,245,967	2,685,570	3,105,960	3,511,624	3,901,981	4,334,166

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 39 Free Net Cash flows Table

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final Value
Net Free Cashflows	-1,598,203	136,044	342,543	425,820	475,375	461,914	479,737	487,060	473,026	458,429	475,123	6,137,011
Discount Factor	1.00	0.71	0.63	0.57	0.51	0.45	0.40	0.36	0.32	0.29	0.26	0.26
Net Present Value for Cash	**-1,598,203**	**96,700**	**217,292**	**241,066**	**240,175**	**208,274**	**193,045**	**174,912**	**151,602**	**131,121**	**121,280**	**1,566,535**

Table below (40) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 40 payback period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	-1,598,203	136,044	342,543	425,820	475,375	461,914	479,737	487,060	473,026	458,429	475,123
Project Value Returned	1,598,203	1,462,159	1,119,616	693,795	218,421	243,493	-723,231	1,210,291	-1,683,317	2,141,74	-2,616,869
Payback Period (Year)	4	1	1	1	1	0	0	0	0	0	0

Table 41 Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%12.05
Net Present Value for Cashflows	1,743,799

Payback Period	4
Internal Rate of Return	%27.85

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Conclusions and Recommendations

Based on the profitability analysis and considering that project inrernal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 1,742,799 Jordan Dinars considering that the project provides 25 Job Opportunitites for the governorate residents.

Table 42 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%27.85	4	12.05%
Revenues declined by 10%	%24.35	5	12.05%
Operating Expenses Increased by 10%	%15.29	7	12.05%